SATURN: Splitting Approach on Tensorized Ubiquitous Regression Networks

Anonymous Author(s)
Affiliation
Address
email

Abstract

Convolutional Neural Networks often contain fully connected layers after their convolution phase, which requires the data to be flattened. Despite empirical success, this flattening ruins the multidimensional structure of the input data and increases the number of parameters, which may cause overfitting. Given the tensor nature of the input data, fully connected layers can be replaced with tensorized low-rank layers, which, besides considering the nature of the data, reduce the number of parameters. However, these layers perform poorly when the approximated rank is too low. To address this issue and further reduce the number of parameters while preserving the quality of the network, we propose a framework that, by using two splitting methods, transforms the multidimensional tensor into a thinner and higher-order tensor and applies tensorized layers on this high-order tensor. This leads to fewer parameters and less computational cost while preserving the quality of the network. In our experiments, evaluated on CIFAR10, MNIST, and Tiny ImageNet-200, we have successfully reduced the classifier parameters of well-known models such as VGG19, ResNet50, and ResNet101 by more than 95%, while also achieving better accuracy and overcoming overfitting. Compared to traditional state-of-the-art models, our approach achieves significant parameter reduction and compression while preserving accuracy.

1 Introduction

Neural Networks, especially those that are applied to images such as *Convolutional Neural Network* (CNNs), often consist of two main parts: One that is responsible to extract features from the input data, and the other part is a *MultiLayer Perceptron* (MLP) which Maps the extracted features to the desired space. Despite the importance of data and their spatial relationship, maintaining their topological structure is often discarded since many known CNNs like ResNet or VGG, seem to work fine; However in cases where these structures are more important than the others such as MRI, It is crucial to adapt the known networks to work without destroying the multi-linear structure of the data. While preserving the natural structure of the data throughout the network is important, a greater drawback from using MLPs is their excessive required number of parameters which not only limit their capabilities in real time and practical problems but also prone them to overfitting.

Most natural datasets can be represented as a Tensor due their multi-linear or multi-modal structure; Images can be considered as a 3-order tensors with width, height and the color channels as their corresponding 3 modes, the same could be considered for videos as 4-order tensor with an extra time mode. In practice, most of the used data in the current machine learning and artificial intelligence tasks can be encoded as tensors, requiring their expansion to the linear algebra.

In machine learning, we usually consider data as vectors and datasets will form a set of vectors (ideally called a matrix). By that definition, many matrix decomposition methods have been invented

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to work on such matrices. Tensor methods are to generalize these methods to a higher dimensional space. Mathematically, tensors, have been widely studied in the past. One excellent example within tensor methods is Tensor Decomposition which can be considered as a prime backbone for most tensorized layers and approximations in deep learning [1] [2], Signal Processing [3] [4], fusion [5], blind source seperation [4] [6], and computer vision [7] .

Considering the new advances in *Deep Neural Networks* (DNNs) and their applications to computer vision such as image recognition, image segmentation, image generation, video generation, vision transformers, etc ...; most architectures still flatten the data before feeding them to their MLPs which not only destroys the natural structure of the data but also increase the number of parameters needed for the model.

Images as the frequent example of data used in machine learning tasks are considered 3-order tensors which require their own algebra to be applied when fed to the MLP. *Fully Connected* (FC) layers used in many architectures are the main culprit behind vectorizing the data and the number of parameters. To alleviate this situation, a better replacement for FC layers is needed which not only preserves the current model's accuracy but also reduces the number of required parameters and does not destroy the structure of data.

One possible working solution to tensorizing FC layers is by employing the low rank approximation of the weight tensor. Some works have been done to address these issues which will be discussed in Section 2 . However, those methods usually perform poorly when the approximated rank is too low which is not ideal for larger models. The cost of these models usually depend on the size of the tensor, our experiment shows that a thinner while higher dimensional reshaped tensor of the original input tensor, can perform much better in lower ranks and by a unique splitting and stacking technique, we preserved the models' accuracy. In this paper, we have provided two splitting approaches which have their own use dependent on the model's task; Using these methods, we have reshaped the tensor to a higher dimensional space where the cost of computation is less and has a better capability to handle lower approximated ranks. Our methods evaluated on CIFAR10, MNIST, and Tiny-ImageNet-200 applied to known CNN models such as ResNet50, ResNet101, and VGG19, as well as a typical CNN achieved similar accuracy in most methods with more than 95 percent less parameters compared to the FC layers and even overcame overfitting in some cases.

2 Related works

Several prior papers address the importance of using tensors and preserving the natural structure, One uses CP decomposition to enhance convolution layers [8]. In [9], tucker decomposition is used for convolution enhancement with fewer parameters. A weight sharing scheme in multi tasking learning is proposed by [10]. In [11], a Tensor-Train decomposition is used as a low rank approximation of the weight tensor. Some uses tensor methods for model compression and acceleration [12].

In general, tensor methods can be widely used in many machine learning and deep learning tasks, especially computer vision [13]. Their use, can be categorized into three main approaches: The first approach compresses the tuned parameters of the pre-trained network while preserving the quality of network with these approximated parameters. So here the new network is not trained and the approximated network could only be used in the test step. Two methods for approximation of the tuned parameters of pertained CNN in each layer based on matrix and tensor decompositions adjusted with clustering of inlet and outlet layers have been proposed [14]. An improvement is reported in [15]. Tensor CP and Tucker decompositions are used to approximate the kernels of pre-trained CNN networks [16] [17] [18]. In [18] a tensor based method to find an approximation of fully connected layer parameters in CNN networks is introduced. Despite the proper implementation results, these methods have the following drawbacks: increase cost of pre-trained networks and finding these compressions is also costly. It also cannot be used in applications such as deep reinforcement learning where the network constantly should be learned. On the other hand, this approach has confined itself to compressing convolutional methods for the test phase, although a deeper question can be asked. Why should we compress an already trained network and not seek for a compact network representation that can be trained from scratch?

The second approach consider the convolutional layers as a base, and approximate or represent them with smaller parameters, to design end-to-end trainable neural network. Different kinds of methods

have been proposed to design end-end learnable networks based on approximation of convolution process with different views: For example, in [19] and [20] , the rank-1 approximation of 3-order tensor filters corresponding to each outlet channel is used. In [21] ,a share factors corresponding to inlet and outlet are considered to reduce the redundancy of parameters. Recently, in [22] by considering the 3-order filters of each outlet channel as rank-k tensor it has been shown that the pixels of each outlet channel is the contraction of this tensor with the corresponding patch of all input channels. Unlike standard convolution, Mobile Nets proposed in [23] separate the spatial filtering and integration steps. For each input channel, only a single spatial filtering is performed, and by different combination of these filtered images, different output channels are obtained.

The third approach do not consider Convolutional structure as a base and try to find other type of layers to work on multidimensional data. One notable recent work is Tensor Regression Networks [1]. They used a *Tensor Regression Layer* (TRL) and *Tensor Contraction Layer* (TCL) to preserve the structure of the data as they are moving across the network while reducing the number of parameters. Their method surprisingly achieved the near similar accuracy as its FC network with more than 65 percent space saving. TRL and TCL will be discussed in section 3, and they are used as the base of our method. Unfortunately, one of the biggest issues with TCL and TRL is their inadequate implementations. They are theoretically required less parameters than their FC versions; However they often perform poorly in terms of the actual execution time. This is one of the biggest limitation on working with tensors since a practical implementation of tensor methods are yet to be made.

The power of tensor regression has been addressed in many prior papers [**?**][**?**] [24] [25]. However, most of these works, usually rely on analytical solution and large tensors. In [26], a non-linear Tensor-Train decomposition is used instead of tucker decomposition. While many works focuses on individual layers, in [27], T-Nets attempts to provide a single high-order, low-rank tensor that jointly capture the full structure of a neural network.

DNNS have been used in many applications and problems, however, many questions still remains in terms of their credibility and why they work so well. DNNs usually contain too many parameters and an important question is whether they require that many parameters. By tensorizing the DNNs, we can address these issues and attempt to better understand the concepts of deep learning. For example, one uses tensor methods as a tool to study CNNs [28], they follow up by studying the power of overlapping architectures in deep learning [29]. Some provide global optimal and optimization for non-convex factorization problems [30]. In [31], they used contraction as a composition operator in *Natural Language Processing* (NLP). Some used tensor methods as a tool to guarantee convergence and consistent learning[32].

Most of the recent works, rely on low rank approximation of the tensor or some tensor decomposition methods as a dimension reduction layer. Despite their success, a fully tensorized end-to-end trainable model requires almost near similar ranks to perform well when compared to their FC competitives. The draw back of using same ranks as the input tensor is the cost. Our method provide a reshaping approach that maintains the accuracy by converting the input tensor into a thinner and higher order tensor that requires fewer parameters and can perform better in lower ranks. To our knowledge, no prior work increases the dimensions of the tensor in order to achieve near similar accuracy while using tensorized layers with lower ranks and fewer parameters.

3 Mathematical background

This paper, requires basic understanding of tensor methods, If readers are already knowledgeable in the matter, they can skip to section 4.

3.1 Notations

For a 3-order tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$, there are 3 modes with respected sizes I_1, I_2, and I_3. Its $\mathcal{T}_{i_1,i_2,i_3}$ **element** is a 0-order tensor (scalar), (i_1, i_2, i_3). **Slices** of its first and second modes are 2-order tensors (matrices) $\mathbf{M}$ and are denoted by two colons, $\mathcal{T}_{:,:,i_3}$. **Fibers** of its first mode are 1-order tensors (vectors) $\mathbf{V}$ and are denoted by a colon, $\mathcal{T}_{:,i_2,i_3}$.

3.2 Tensor folding and unfolding

Tensor folding and unfolding are fundamental operations in tensor analysis, allowing for the manipulation and transformation of tensors into different dimensional structures. Tensor unfolding, also known as matricization, converts a higher-order tensor into a matrix or a vector. Let $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \ldots \times I_N}$ be an N-order tensor. In our case, tensor folding folds an unfolded tensor back.

Mode-n unfolding : Unfolding $\mathcal{T}$ along mode-n results in a matrix representation of mode-n. This operation is denoted as $T_{[n]}$. The unfolding is achieved by rearranging the entries of $\mathcal{T}$ into a matrix such that the mode-n indices become the rows of the matrix. The mapping from tensor indices to matrix indices is given by:

$$\begin{aligned} &(i_1, \ldots, i_n, \ldots, i_N) \longrightarrow (i_n, r) \\ &\text{where } r = \sum_{\substack{p=1 \\ p \neq n}}^{N} i_p \times \prod_{\substack{l=p+1 \\ l \neq n}}^{N} I_l \end{aligned} \tag{1}$$

Each row of the resulting matrix corresponds to a different combination of indices for the remaining modes.

Vectorization : Similar to mode-n unfolding, in tensor vectorization we map each element within the tensor to a element j of $Vec(\mathcal{T})$:

$$\begin{aligned} &(i_1, \ldots, i_n, \ldots, i_N) \longrightarrow (r) \\ &\text{where } r = \sum_{p=1}^{N} i_p \times \prod_{l=p+1}^{N} I_l \end{aligned} \tag{2}$$

3.3 Tensor contraction

Tensor contraction is a fundamental operation that allows for the multiplication and simplification of tensors and multiplies elements of two tensors and provides a way to combine tensors by summing over shared indices.

Tensor n-mode product : Tensor n-mode product, is a tensor operation that allows for the multiplication of a tensor with a matrix along a specific mode. Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and a matrix $\mathbf{M} \in \mathbb{R}^{J \times I_n}$, the n-mode product of $\mathcal{T}$ and $\mathbf{M}$, denoted as $\mathcal{T} \times_n \mathbf{M}$, results in a tensor of size $\mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times J \times I_{n+1} \times \cdots \times I_N}$. The n-mode product can be expressed mathematically as:

$$(\mathcal{T} \times_n \mathbf{M})_{i_1, \ldots, i_{n-1}, j, i_{n+1}, \ldots, i_N} = \sum_{i_n=0}^{I_n} \mathcal{T}_{i_1, \ldots, i_n, \ldots, i_N} \mathbf{M}_{j, i_n} \tag{3}$$

where $i_1, \ldots, i_{n-1}, j, i_{n+1}, \ldots, i_N$ are the indices of the resulting tensor, and i_n is the summation index. Alternatively, the n-mode product can be expressed using the tensor unfolding operation along the n-th mode, denoted as $\mathbf{T}_{(n)}$. In this case, the n-mode product can be written as:

$$\mathcal{T} \times_n \mathbf{M} = \mathbf{M}\mathbf{T}_{[n]} \tag{4}$$

Generalized inner product : For two tensors $\mathcal{X} \in \mathbb{R}^{I_x \times I_1 \times I_2 \times \cdots \times I_N}$ and $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N \times I_y}$ sharing N modes of the same size, a generalized inner product along the N last modes of X and Y is defined as:

$$\langle \mathcal{X}, \mathcal{Y} \rangle_N = \sum_{i_1=0}^{I_1 - 1} \cdots \sum_{i_N=0}^{I_N - 1} \mathcal{X}_{:, i_1, i_2, \ldots, i_N} \mathcal{Y}_{i_1, i_2, \ldots, i_N, :} \tag{5}$$

where $\langle \mathcal{X}, \mathcal{Y} \rangle_N \in \mathbb{R}^{I_x \times I_y}$.

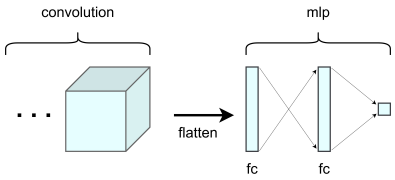


Figure 1: A simple MLP after some convolution layers.

3.4 Tucker decomposition

The Tucker decomposition is a powerful method in tensor algebra that breaks down a tensor into a core tensor multiplied by matrices along each mode. In the Tucker decomposition, a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ is represented as the product of a core tensor $\mathcal{G} \in \mathbb{R}^{J_1 \times J_2 \times \cdots \times J_N}$ and a set of factor matrices $\mathbf{A}^{(1)} \in \mathbb{R}^{I_1 \times J_1}, \mathbf{A}^{(2)} \in \mathbb{R}^{I_2 \times J_2}, \ldots, \mathbf{A}^{(N)} \in \mathbb{R}^{I_N \times J_N}$ along each mode. Mathematically, the Tucker decomposition can be expressed as:

$$\mathcal{T} = \mathcal{G} \times_1 \mathbf{A}^{(1)} \times_2 \mathbf{A}^{(2)} \times \cdots \times_N \mathbf{A}^{(N)} \tag{6}$$

where $\times_n$ denotes the n-mode product. The Tucker decomposition can be computed using the mode-n unfolding of the tensor, denoted as $\mathbf{T}_{[n]}$. By reshaping the tensor along each mode, the Tucker decomposition can be written as:

$$\mathbf{T}[n] = \mathbf{A}^{(n)} \mathbf{G}[n] \left(\mathbf{A}^{(1)} \otimes \cdots \otimes \mathbf{A}^{(n-1)} \otimes \mathbf{A}^{(n+1)} \otimes \cdots \otimes \mathbf{A}^{(N)} \right)^T \tag{7}$$

where $\mathbf{G}_{[n]}$ represents the unfolding of the core tensor along mode n.

3.5 Tensor regression networks

The combination of tensor regression and tensor contractions has been used in an end-to-end trainable model by Kossaifi [1], as discussed in section 2 Given a MLP, there are two ways to interpret FC layers: One is to consider them as a feature extraction layer where they map a latent or feather space to another latent space, on another hand, we can consider them as regression layers which are basically the classifier layers and given the purpose of the network, they can be used interchangeably. In figure 1 a very simple MLP that is being used after some convolution layers is provided. In this case, the first FC layer can be considered a dimension reduction layer while the second FC layer maps the feature space to the target space. In the following text, we will discuss to main approaches to replace such FC layers:

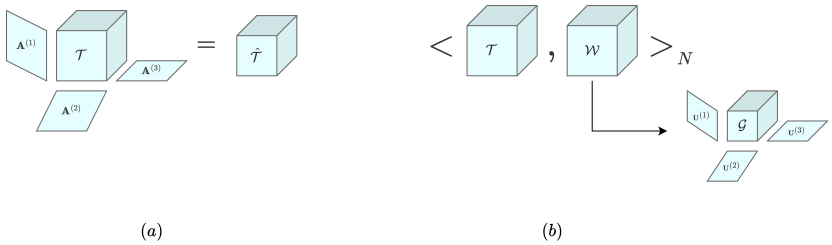
3.5.1 Tensor contraction layer

Tensor Contraction Layer (TCL) is a form of feature extraction and could also be used for embedding. The prime idea behind TCL is Tucker decomposition; Via multiplying matrices to each mode of the input tensor, we can obtain a denser tensor (dependent of the shape of the matrices). Figure 2 demonstrate the TCL process on a 3-order tensor.

Considering an input tensor $\mathcal{X}$ of size $(S_0, I_1, \ldots, I_N)$ and a size $(R_1, \ldots, R_N)$ TCL, where S_0 is the size of the batch and (I_i, R_i) is the shape of the matrix applied to mode i of the tensor; The number of required parameters is calculated as :

$$\sum_{i=1}^{N} R_i \times I_i \tag{8}$$

And the resulting tensor is of size $(S_0, R_1, \ldots, R_N)$. Compared to the FC version where the simple regression between these two tensors would require at least $\prod_{i=1}^{N} I_i \times R_i$ parameters.



(a) (b)

Figure 2: Tensor regression networks, replacement for FC layers. In (a), A simple TCL is applied; Each mode i of tensor $\mathcal{T}$ is multiplied by factor $\mathbf{A}^{(i)}$. In (b), A TRL is applied; Generalized inner product of input tensor $\mathcal{T}$ and weight tensor $\mathcal{W}$ is calculated as the output tensor while considering a low rank approximation of weight tensor $\mathcal{W}$ in a tucker decomposed form.

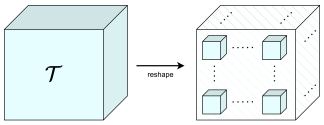


Figure 3: An outline of reshaping to higher orders tensor.

3.5.2 TRL

Tensor Regression Layers (TRL) is a replacement for the FC regression layers. A simple regression that is the foundation of a MLP is a trivial generalized inner product of $\mathcal{W}$ (Weight Tensor) with the $\mathcal{X}$ (Input tensor). The idea behind TRL is to reconstruct W from a low rank approximation of itself via tucker decomposition. In figure 2, a TRL application in a MLP is provided. For the sake of simplicity, we considered a binary classification problem where the output of the inner product is a scaler. Similarly, if the desired output is a vector or even a tensor, $\mathcal{W}$ will be a higher dimensional tensor itself to generate each mode of the output.

Considering an input tensor $\mathcal{X}$ of size $(S_0, I_1, \ldots, I_N)$, an output tensor $\mathcal{Y}$ of size $S_0 \times O$, and a size $(R_1, \ldots, R_N, R_{N+1})$ TRL with bias b of size O such that $\mathcal{Y} = \langle X, W \rangle_N + b$; The number of required parameters can be calculated as :

$$\prod_{i=1}^{N+1} R_i + \left(\sum_{i=1}^{N} R_i \times I_i + R_{N+1} \times O \right) \tag{9}$$

Compared to the FC version where the number of parameters is $O \times \prod_{i=1}^{N} I_i$.

4 Splitting approach

Impressed by the vision transformers patching and down sampling methods, we can transform our input tensor into a thinner and higher dimensional tensor that not only reduces the number of parameters for a TRL or TCL layer but also maintain the accuracy. However, not any reshaping scheme to arbitrary higher orders and dimensions guarantee that the multi-linear structure is preserved and the accuracy is maintained With a proper splitting scheme, the newly made tenser has better physical interpretation than the flattened data. We introduced two splitting schemes, both of which are useful and could be used differently according to the network architect. Figure 3 demonstrate the general splitting outline. Given a 3-order tensor, if we split across each mode, the output is a 6-order tensor where the first 3 modes correspond the split index, and the next 3 modes are the data in each split.

4.1 Attached Splitting

In attached splitting, every split is continues, that is if mode I_i of the input tensor $\mathcal{X}$ with length n is to be split into 2 parts, then the split indices are $(0, \frac{n}{2})$ and $(\frac{n}{2}, n)$. Figure 4 demonstrate the splitting process on a matrix.

Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ and split sizes $(L_1, \ldots, L_N)$, attached mapping of elements is done via:

$$\mathcal{T}[i_1, \ldots, i_N] \longrightarrow \mathcal{T}_a \left[\lfloor \frac{L_1 \times i_1}{I_1} \rfloor, \ldots, \lfloor \frac{L_N \times i_N}{I_N} \rfloor, i_1 \bmod \frac{I_1}{L_1}, \ldots, i_N \bmod \frac{I_N}{L_N}, \right] \tag{10}$$

where $\mathcal{T}_a \in \mathbb{R}^{L_1 \times \cdots \times L_N \times \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}}$.

4.2 Compressed Splitting

In compressed splitting, we incorporate the idea behind down sampling, that is if mode I_i of input tensor $\mathcal{X}$ with length n is to be split into 2 part , different to the attached splitting, the indices are even-odd combinations; That is $\{t | t \in (0, n) \wedge (t \bmod 2 = 0)\}$ and $\{t | t \in (0, n) \wedge (t \bmod 2 = 1)\}$.

Given a tensor $\mathcal{T} \in \mathbb{R}^{I_1 \times \cdots \times I_N}$ and split sizes $(L_1, \ldots, L_N)$, compressed mapping of elements is done via:

$$\mathcal{T}[i_1, \ldots, i_N] \longrightarrow \mathcal{T}_c \left[i_1 \bmod L_1, \ldots, i_N \bmod L_N, \lfloor \frac{i_1}{L_1} \rfloor, \ldots, \lfloor \frac{i_N}{L_N} \rfloor, \right] \tag{11}$$

where $\mathcal{T}_c \in \mathbb{R}^{L_1 \times \cdots \times L_N \times \frac{I_1}{L_1} \times \cdots \times \frac{I_N}{L_N}}$.

4.3 Method Analysis

Considering an input tensor $\mathcal{X}$ of size $(S_0, I_1, \ldots, I_N)$, either reshaping would result in the tensor $\hat{\mathcal{X}}$ of size $(S_0, L_1, \ldots, L_N, \frac{I_1}{L_1}, \ldots, \frac{I_N}{L_N})$. An example of each reshaping approach is done in figure 5 on a sample image.

In case of a $(r_1 \ldots, r_N, R_i, \ldots, R_N)$ TCL, Since L_is are usually small, the number of parameters required for them $\sum_{i=0}^{N} L_i \times r_i$ is also small; Similarly, $\frac{I_i}{L_i}$ is smaller than I_i so the new desired R_is is smaller than the previous R_is and the total number of parameters can be calculated as:

$$\sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} \tag{12}$$

which is ideally less than equation 8.

In case of $(r_1, \ldots, r_N, R_1, \ldots, R_N, R_{N+1})$ TRL, the same logic as TCL could also be applied. The total number of parameters in a new TRL is calculated as:

$$\left(\prod_{i=1}^{N} r_i \prod_{i=1}^{N+1} R_i \right) + \left(\sum_{i=1}^{N} L_i \times r_i + \sum_{i=1}^{N} R_i \times \frac{I_i}{L_i} + R_{N+1} \times O \right) \tag{13}$$

which is ideally less than equation 9.

5 Experiments

6 Conclusion

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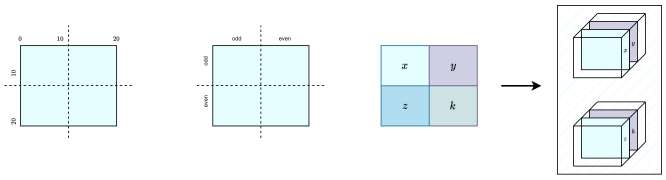

(a) (b) (c)

Figure 4: (a) attached splitting of a 2-order tensor. (b) compressed splitting of a 2-order tensor. (c) demonstration of stacking split chunks of a 2-order tensor into a 4-order tensor.

(a) (b) (c)

Figure 5: (a) original sample image. (b) reshaped sample image using attached approach. (c) reshaped sample image using compressed approach.

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